CRIME DATA ANALYSIS

Los Angeles Crime Data, 2020 – Present

An End-to-End Data Analytics Report

Scope of Work

Data Cleaning • Exploratory Data Analysis • Database Engineering (PostgreSQL)
Interactive Business Intelligence Dashboard (Power BI)

Prepared using: Python (Pandas, NumPy, Matplotlib, Seaborn) · PostgreSQL · Power BI
Dataset: Crime_Data_from_2020_to_Present.csv (1,005,198 records · 24 fields)
Report Date: August 2026

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1. Executive Summary

This report presents an end-to-end analysis of the "Crime Data from 2020 to Present" dataset, a record-level extract of crime incidents reported to the Los Angeles Police Department (LAPD). The project covered the complete analytics lifecycle: data ingestion and quality assessment, cleaning and preprocessing in Python, exploratory data analysis (EDA), storage in a relational database (PostgreSQL), and the construction of an interactive, multi-page Power BI dashboard for ongoing self-service reporting.

The source file contains 1,005,198 individual crime records across 24 original fields, spanning report dates from 2020 through early 2025. After profiling and cleaning, a focused analytical table of 1,005,198 records and 15 fields was produced and loaded into PostgreSQL, alongside a copy of the untouched raw table for audit and traceability. This cleaned table underpins both the Python-based EDA in this report and the Power BI dashboard delivered alongside it.

Key findings show that crime is heavily concentrated around the midday hour (12:00 PM) and again during the early-evening commute window (5 PM – 8 PM); that a small group of divisions — Central, 77th Street, Pacific, Southwest and Hollywood — account for a disproportionate share of citywide incidents; and that theft- and vehicle-related offenses (Vehicle – Stolen, Battery – Simple Assault, Burglary from Vehicle) dominate the citywide crime mix. Victim-age analysis is directionally useful but is affected by a data-quality artifact discussed in Section 8 (Limitations), and should be read with that caveat in mind.

The remainder of this report documents the methodology, presents the detailed findings with supporting visuals, describes the database and dashboard architecture, and closes with limitations and recommendations for operational use and future analytical work.

2. Introduction & Objectives

Public safety agencies generate large volumes of incident-level data that, if properly structured and analyzed, can inform resource allocation, patrol scheduling, and community safety communication. The objective of this project was to convert a raw, semi-structured crime-incident export into a clean, analysis-ready dataset and to surface actionable patterns in when, where, and to whom crime occurs.

2.1 Objectives

- **Data Quality:** Profile the raw dataset, quantify missingness and duplication, and apply a defensible, documented cleaning process.
- **Exploratory Analysis:** Identify temporal patterns (hour-of-day), spatial patterns (division/area), and demographic patterns (victim age and sex) in reported crime.
- **Data Engineering:** Persist both the raw and cleaned datasets in a PostgreSQL relational database to support repeatable querying and downstream BI tools.
- **Business Intelligence:** Deliver an interactive Power BI dashboard that allows non-technical stakeholders to filter, drill down, and monitor crime trends without writing code.

2.2 Toolset

The project was implemented using Python 3 (pandas, NumPy, Matplotlib, Seaborn) in a Jupyter Notebook for data preparation and EDA; PostgreSQL as the relational data store, accessed via SQLAlchemy and psycopg2; and Microsoft Power BI Desktop for dashboarding, connected live to the PostgreSQL database.

3. Dataset Overview

The source file, Crime_Data_from_2020_to_Present.csv, is consistent in structure with the LAPD's publicly released crime-incident export. It captures one row per reported crime incident, including when and where it was reported and occurred, how it was classified, and (where available) victim demographic details and the eventual case status.

3.1 Structure

Property	Value
Total records (rows)	1,005,198
Original fields (columns)	24
Fields retained after cleaning	15
Grain	One row per reported crime incident (DR_NO)
Primary key	DR_NO (division of records number) — confirmed unique
Full row duplicates	0
Date range covered	Report dates from 2020 through early 2025

3.2 Key Original Fields

Field	Description
DR_NO	Unique division of records number (incident identifier)
Date Rptd / DATE OCC	Date the crime was reported / date it occurred
TIME OCC	24-hour time of occurrence (military format)
AREA / AREA NAME	LAPD patrol division code and name (21 divisions)
Crm Cd / Crm Cd Desc	Numeric crime code and its text description
Vict Age / Vict Sex / Vict Descent	Victim age, sex, and descent (self-reported / recorded)
Premis Cd / Premis Desc	Code and description of the premises where the crime occurred
Status / Status Desc	Case status (e.g., Invest Cont, Adult Arrest, Juv Arrest)
LOCATION / Cross Street / LAT / LON	Address-level and geographic coordinates (anonymized to block level)

4. Data Cleaning & Preprocessing

Data preparation was performed in Python using pandas, following a standard profile → clean → validate workflow. All transformations were scripted in the accompanying Jupyter notebook, ensuring the process is fully reproducible.

4.1 Data Quality Profile (Raw Data)

An initial null-value and duplicate audit was run across all 24 original columns. Two integrity checks (full-row duplicates and DR_NO primary-key duplicates) both returned zero, confirming each record represents a distinct, uniquely identified incident. Missingness, however, was uneven across columns:

Field	Missing Values	Missing %
Crm Cd 2	936,039	93.12%
Cross Street	850,955	84.66%
Mocodes	151,760	15.10%
Vict Descent	144,794	14.40%
Vict Sex	144,782	14.40%
Premis Desc	588	0.06%
Premis Cd	16	<0.01%
Crm Cd 1 / Status	11 / 1	<0.01%

4.2 Cleaning Steps Applied

- **1. Column-level pruning:** Columns missing more than 50% of their values (Crm Cd 2, Cross Street) were dropped programmatically using a 0.5 non-null threshold, since they were too sparse to analyze reliably and represent optional/secondary fields (a second crime code, or a cross-street only recorded for intersection-based incidents).
- **2. Scope reduction:** Additional fields not required for the planned temporal, spatial, and demographic analysis — Part 1-2, Mocodes, Vict Descent, Premis Cd, Status, Crm Cd 1, LOCATION, LAT and LON — were dropped to keep the working table focused. (The full, untouched raw table was preserved separately for any downstream geographic or modus-operandi analysis — see Section 8.)
- **3. Victim age imputation:** Vict Age contained a large number of zero values, used in the source system as a placeholder where no individual victim age applies (e.g., stolen-vehicle or property-only offenses) or where age was not captured. These zeros were replaced with the mean age of records with a genuine (non-zero) age, 39.5 years, and the column was then cast to an integer type.
- **4. Categorical imputation:** Missing Vict Sex and Premis Desc values were filled with the statistical mode (most frequent category) of each column, a standard approach for low-cardinality categorical fields with limited missingness.

- **5. Feature engineering:** TIME OCC was parsed to extract an Hour (0–23) field, and Vict Age was binned into eight analytical Age Group buckets (0–10, 11–20, …, 71+) to support demographic aggregation.

Following these steps, the working table (df1) was reduced to 15 fields — DR_NO, Date Rptd, DATE OCC, TIME OCC, AREA, AREA NAME, Rpt Dist No, Crm Cd, Crm Cd Desc, Vict Age, Vict Sex, Premis Desc, Status Desc, Hour and Age Group — with zero remaining nulls, verified by a final isnull().sum() check returning all zeros.

5. Exploratory Data Analysis

With a clean, feature-enriched dataset in place, five analytical lenses were applied: time-of-day, crime-type-by-time, crime-type-by-area, area-level volume, and victim demographics. Each is presented below with its supporting visual, generated directly from the cleaned data.

5.1 Crime Occurrence by Time of Day

Aggregating all 1,005,198 incidents by hour of occurrence reveals a distinctive daily rhythm rather than a flat distribution.

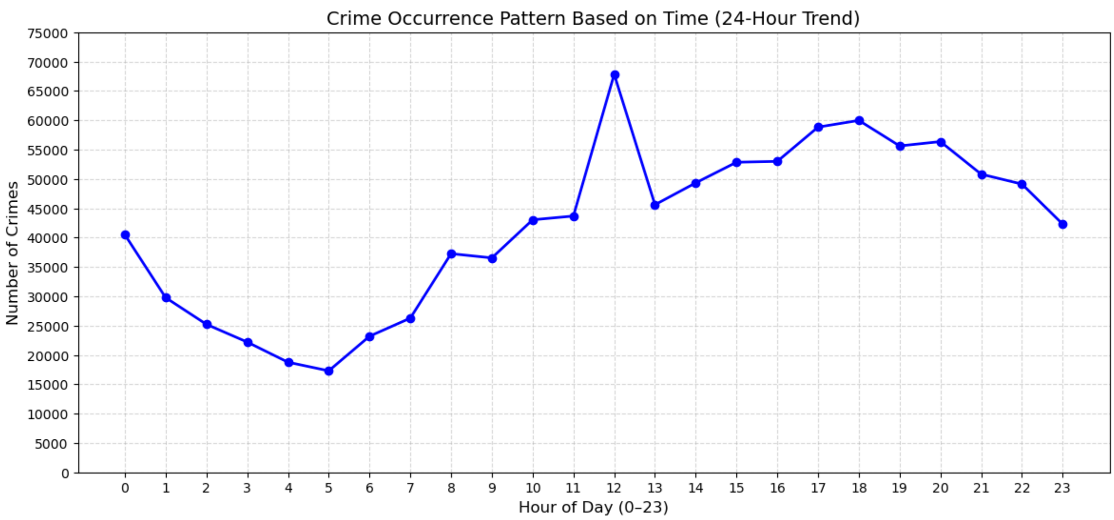

Figure 1. Crime occurrence pattern by hour of day (24-hour trend), all incidents.

- **Midday peak:** Incidents spike sharply at 12:00 (noon), reaching roughly 68,000 — the single highest hourly total in the dataset, and markedly above both neighboring hours.
- **Overnight trough:** Volumes bottom out between 04:00 and 05:00, at roughly 17,000–19,000 incidents — about a quarter of the noon peak.
- **Evening rise:** A second, broader elevated band runs from 15:00 through 20:00, generally 53,000–60,000 incidents per hour, consistent with increased public activity after the workday.

Note: the sharpness of the noon spike is larger than a purely behavioral pattern would typically produce, and likely reflects, in part, a data-entry convention where 12:00 is used as a default/rounded timestamp when the exact time of occurrence is unknown or estimated. This is discussed further in Section 8.

5.2 Crime Type vs. Hour of Occurrence

To understand which offense types drive the time-of-day pattern, the top 20 crime types (by volume) were cross-tabulated against hour of occurrence.

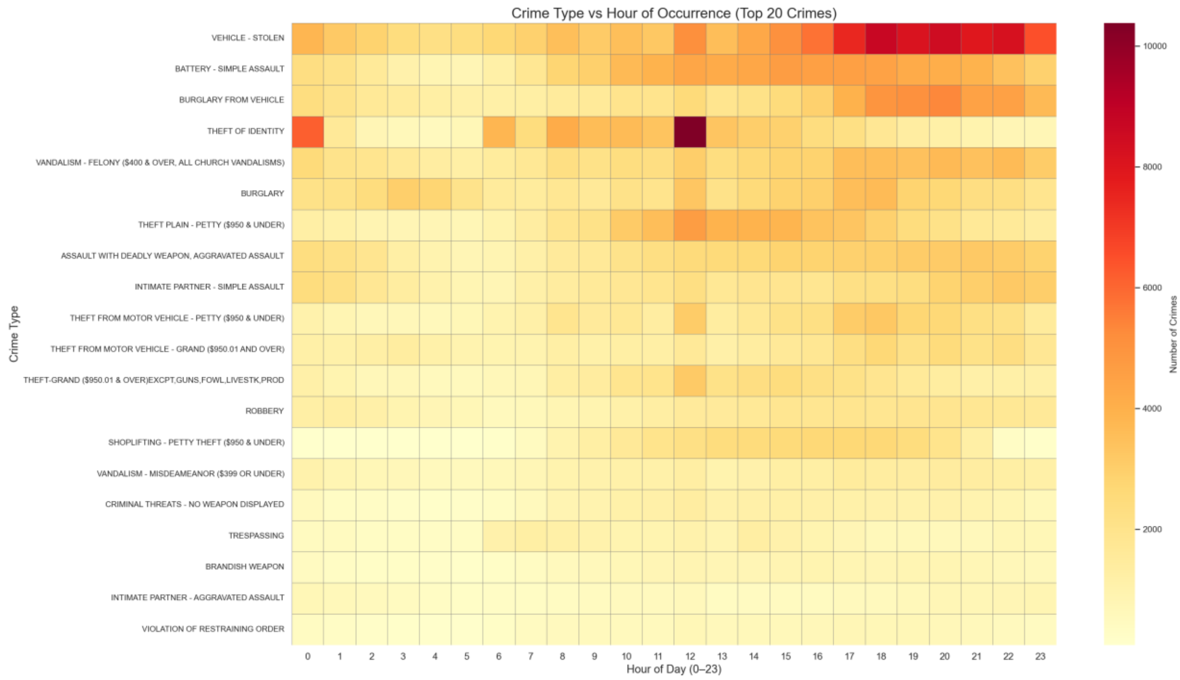

Figure 2. Heatmap of the top 20 crime types by hour of occurrence.

- **Vehicle – Stolen:** Shows the strongest and most sustained hourly intensity of any offense, building steadily from mid-afternoon and staying elevated from roughly 15:00 through midnight — consistent with vehicles being reported stolen after being parked and left unattended for extended periods.
- **Theft of Identity:** Displays a very different, bursty pattern — a pronounced concentration at hour 0 (midnight) and an especially dark cell at hour 12, rather than a smooth daily curve. This is characteristic of a non-time-sensitive offense being logged with a default timestamp (00:00 or 12:00) rather than a true occurrence time, since identity-theft discovery dates rarely map to a precise hour.
- **Battery – Simple Assault and Burglary from Vehicle:** Both trend upward through the afternoon and evening, broadly tracking the citywide curve from Figure 1.

5.3 Crime Type vs. Area of Occurrence

The same top-20 crime types were then cross-tabulated against the 21 LAPD geographic areas to identify area-specific offense concentrations.

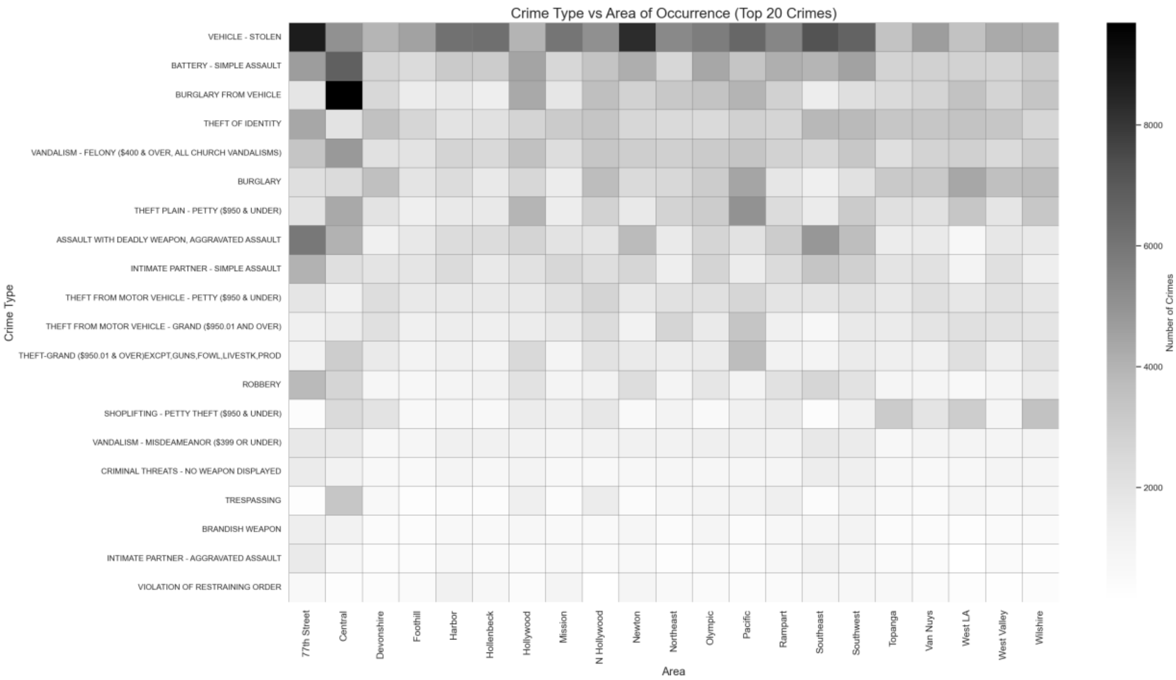

Figure 3. Heatmap of the top 20 crime types by patrol area.

- **Central:** Stands out with the darkest cells for both Vehicle – Stolen and, especially, Burglary from Vehicle — the single most concentrated crime-type/area combination in the dataset.
- **77th Street:** Shows the highest concentration of Assault with Deadly Weapon / Aggravated Assault of any division, indicating a distinct violent-crime profile relative to other areas.
- **Southeast and Pacific:** Register elevated Theft Plain – Petty and general theft-type offenses relative to most other divisions.

5.4 Crime Volume by Area

Ranking the 21 LAPD divisions by total incident count highlights where citywide crime volume concentrates geographically.

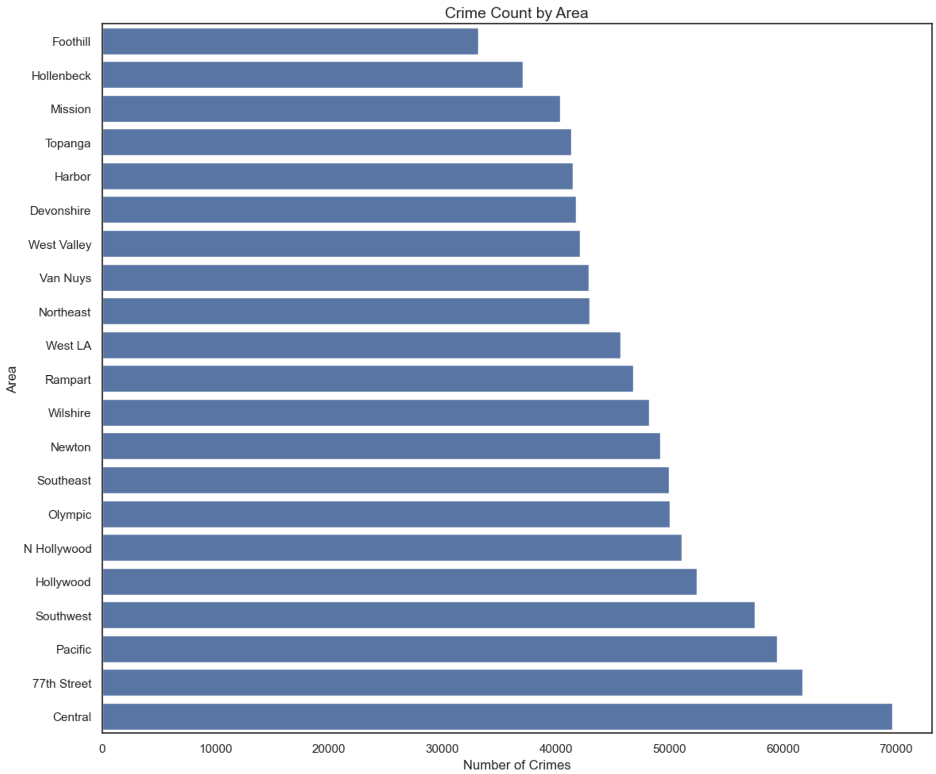

Figure 4. Total crime count by patrol area, ranked ascending.

Rank	Area	Crime Count
1 (highest)	Central	69,674
2	77th Street	61,763
3	Pacific	59,520
4	Southwest	57,511
5	Hollywood	52,430
…	…	…
19	Hollenbeck	37,095
20	Mission	40,365
21 (lowest)	Foothill	33,136

The gap between the busiest division (Central, 69,674 incidents) and the quietest (Foothill, 33,136) is more than 2:1, underscoring that crime volume is not evenly distributed across the city and that a resourcing strategy calibrated to citywide averages would under-serve the highest-volume divisions.

5.5 Crime Volume by Area and Time

Combining the area and time dimensions into a single heatmap shows how the citywide midday spike (Figure 1) plays out differently across divisions.

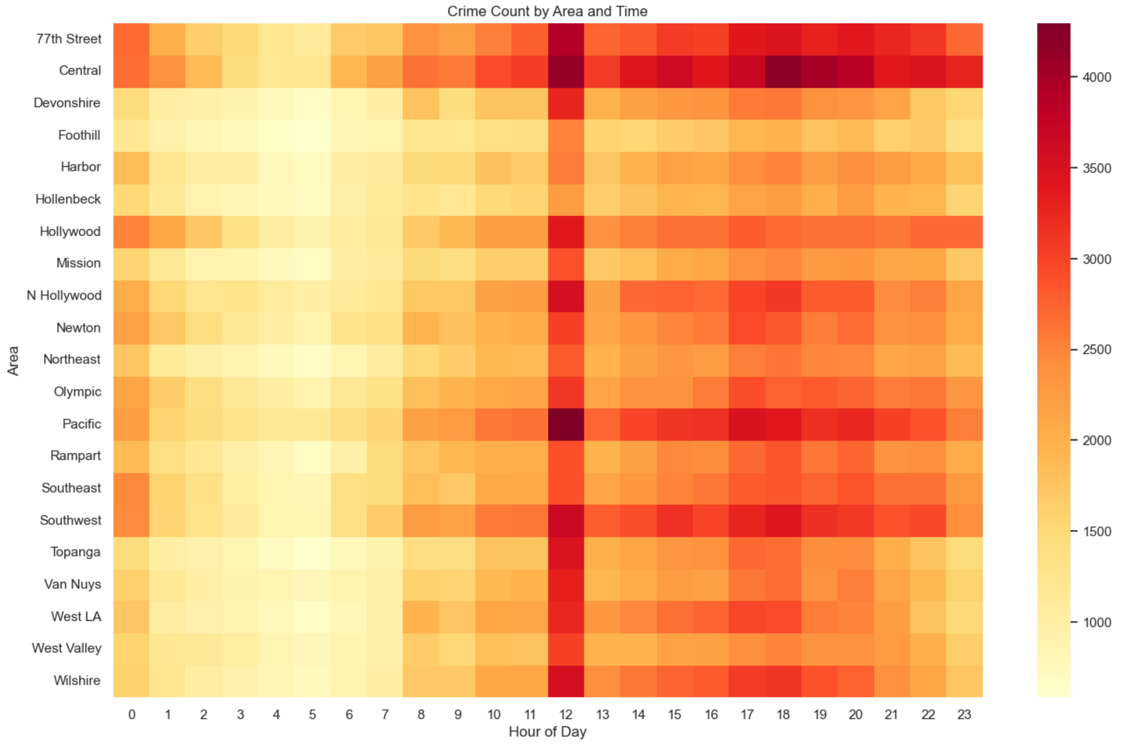

Figure 5. Crime count by area and hour of day.

- **Central and Pacific:** Show the darkest single cells in the entire matrix at the 12:00 hour, confirming these two divisions are the largest contributors to the citywide noon spike.
- **Sustained evening activity:** Central, 77th Street, Pacific, Southwest and Hollywood all remain in the higher-intensity color band from midday through late evening (roughly 12:00–22:00), while lower-volume divisions such as Foothill, Hollenbeck and Mission stay in the pale end of the scale throughout the day.

5.6 Victim Age Distribution

Victim ages (post-imputation, see Section 4.2) were grouped into eight bands to profile who is most frequently recorded as a victim.

Age Group	Record Count	Share of Total
0–10	5,286	0.5%
11–20	48,340	4.8%
21–30	190,140	18.9%
31–40	457,964	45.6%
41–50	126,870	12.6%
51–60	93,657	9.3%
61–70	54,737	5.4%
71+	28,067	2.8%

The 31–40 band accounts for 45.6% of all records — far more than would be expected from a natural age distribution. This is a direct consequence of the mean-imputation step in Section 4.2: every record whose original Vict Age was 0 (no individual victim, or age not captured) was assigned the dataset mean of 39.5, which falls inside the 31–40 bucket. This concentration should be read as a data-processing artifact rather than a genuine demographic finding; see Section 8 for a recommended alternative treatment.

5.7 Victim Age vs. Crime Type

Cross-tabulating victim age group against the 15 most frequent crime types shows where the age concentration in Section 5.6 originates.

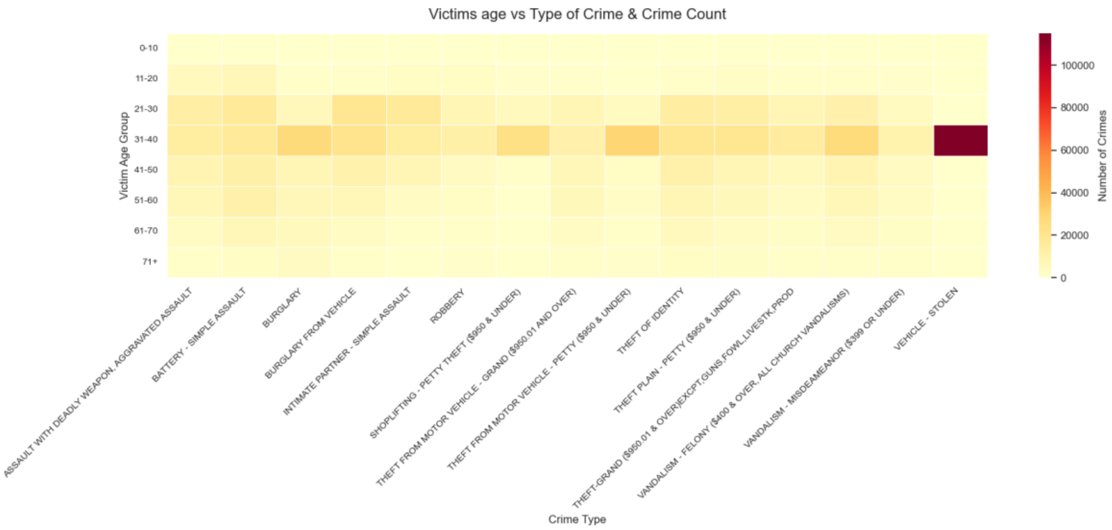

Figure 6. Heatmap of victim age group against the top 15 crime types.

The single dominant cell in the entire matrix is Vehicle – Stolen within the 31–40 age band, far exceeding every other combination. Vehicle-theft incidents typically have no individual victim and would originally carry Vict Age = 0; after imputation, all of these records were pushed into the same 31–40 bucket at the mean age, producing this outsized cell. This reinforces the Section 5.6 caveat: age-based conclusions are most reliable for person-directed offenses (assault, robbery) and should be treated cautiously for property offenses like vehicle theft and burglary.

6. Data Engineering: PostgreSQL Integration

To make the dataset queryable at scale and to serve as a live source for the Power BI dashboard, both the raw and the cleaned datasets were loaded into a PostgreSQL database named crime_records, using SQLAlchemy's create_engine with the psycopg2 driver.

Table Name	Source DataFrame	Contents
crime_data	df1 (cleaned)	15-field analysis-ready table used for EDA and the Power BI dashboard
crime_data_actual	df (raw)	Full 24-field, unmodified source data, preserved for audit and future re-processing

Loading both the raw and cleaned tables (rather than overwriting the source) preserves a complete audit trail: any cleaning decision made in Section 4 can be revisited or re-derived directly from crime_data_actual without needing to re-import the original CSV file.

Recommendation: the database connection string in the current notebook embeds the username and password directly in plain text. For any shared or production use, these credentials should be moved to environment variables or a secrets manager (e.g., a .env file excluded from version control, or a cloud secrets service) rather than being hardcoded in notebook source code.

7. Power BI Dashboard

An interactive, five-page Power BI dashboard (crime_data_analysis_dashboard.pbix) was built on top of the crime_data PostgreSQL table, giving non-technical stakeholders self-service access to the same patterns explored in Section 5, with the ability to filter and drill down live.

Every page shares a consistent set of cross-filtering slicers — typically Year, Month, Area, Crime Type, and, where relevant, Victim Sex, Age Group, Report District, or Case Status — so a stakeholder can, for example, select a single division and instantly see every KPI card and chart on the page recalculate.

7.1 Page 1 — Crime Overview

The landing page gives a citywide, at-a-glance summary.

- KPI cards: Total Crimes, Total Crime Types, Average Victim Age, Most Frequent Crime, Total Areas.
- Crime Trend Over Time (Yearly) — line chart of incident volume by month/year.
- Crimes by Area — bar chart ranking all 21 divisions.
- Crimes by Crime Type — bar chart of the leading offense categories.
- Crime Trend Over Time (Hourly) — line chart re-creating the Figure 1 pattern interactively.
- Victim Gender Distribution — donut chart.

7.2 Page 2 — Area & Crime Type Analysis

A deeper drill-down page for spatial and categorical analysis.

- KPI cards: Top Crime Area, Total Crime Categories, Least Frequent Crime, Most Frequent Crime.
- Top 10 Crime Types and Crimes by Premises — bar charts.
- Crime Status Distribution — donut chart (e.g., Invest Cont, Adult Arrest).
- Crimes per Area — pivot table for precise figures.
- No. of Crime Types per Area — bar chart showing offense diversity by division.

7.3 Page 3 — Victim Demographics

Focused entirely on victim-level attributes.

- KPI cards: Average Victim Age, Youngest Victim Age, Oldest Victim Age, Male Victims, Female Victims.
- Victim Age Group Distribution — column chart.
- Victim Gender Distribution — donut chart.
- Average Victim Age by Area and by Crime Type — bar charts.
- Victim Gender by Crime Type — 100% stacked bar chart.

7.4 Page 4 — Temporal Patterns

Dedicated to time-based drill-down beyond the hourly view on Page 1.

- KPI cards: Peak Hour of Crime, Peak Month of Crime, Most Frequent Crime Day, Average Crimes per Day.
- Average Crimes per Month of Year — column chart.
- Average Crimes per Day of Week — column chart.
- Crimes by Time of Day (categorical Morning/Afternoon/Evening/Night bands) — column chart.

7.5 Page 5 — Detailed Records

A record-level, filterable data table (DR_NO, Date Rptd, DATE OCC, TIME OCC, Area, Report District, and related fields) that lets analysts inspect or export the underlying incidents behind any KPI or chart selection made on the previous pages.

8. Limitations & Data Quality Notes

The following caveats should be considered when using this analysis or dashboard for operational decision-making:

- **Reported crime, not actual crime:** The dataset reflects incidents reported to and recorded by LAPD; it does not capture unreported crime, so true incidence may differ from reported volume, particularly for offenses with low reporting rates.
- **Victim age imputation artifact:** As detailed in Sections 4.2, 5.6 and 5.7, zero/missing victim ages were imputed with the overall mean (39.5), which materially inflates the 31–40 age band and the apparent link between that band and property crimes such as Vehicle – Stolen. A future iteration should either exclude Vict Age = 0 records from age-based analysis entirely, or add an explicit “not applicable / unknown” category rather than imputing a numeric age.
- **Default/rounded timestamps:** The sharp spikes at hour 0 and hour 12 in Figures 1 and 2 (most visible for Theft of Identity) are consistent with placeholder timestamps used when an exact occurrence time is unknown, rather than a genuine behavioral pattern. Hourly conclusions should therefore be treated as directionally indicative rather than precise.

- **Dropped fields:** Crm Cd 2 (93.1% missing) and Cross Street (84.7% missing) were dropped for the analytical table; a second-crime-code analysis or intersection-level geocoding would require reintroducing them from the raw crime_data_actual table.
- **Geospatial detail removed:** LAT, LON and LOCATION were excluded from the cleaned table used for this analysis. Area-level (division) analysis was used as a proxy for geography; true hot-spot mapping would require rejoining these fields from the raw table.
- **Victim Descent excluded:** Vict Descent was dropped from the analytical table and was not profiled in this report; it remains available in the raw table for a future demographic deep-dive.
- **Credential handling:** The current notebook stores database credentials as plain-text variables, which is not recommended practice outside of a local, single-user environment (see Section 6).

9. Recommendations

9.1 Operational

- **Calibrate patrol deployment to the two peak windows:** the sharp midday spike (≈12:00) and the sustained evening band (≈15:00–20:00), rather than a flat 24-hour allocation.
- **Prioritize the top five divisions:** Central, 77th Street, Pacific, Southwest and Hollywood together account for a disproportionate share of citywide volume and warrant a resourcing model calibrated above the citywide average.
- **Target vehicle-theft prevention:** Vehicle – Stolen is the leading or near-leading offense in the top divisions and shows a distinct evening/overnight intensity band — a strong candidate for a focused deterrence or public-awareness campaign.

9.2 Data & Analytics

- **Fix the age-imputation approach:** Replace mean-imputation of Vict Age = 0 with an explicit “not applicable” category, or exclude property-only offenses from victim-age analysis, before this field is used in further demographic reporting.
- **Audit timestamp defaults:** Work with LAPD data-entry guidance (or the records-management system) to confirm whether 00:00/12:00 are placeholder times, and if so, flag or exclude them from hour-of-day analysis.
- **Reintroduce geography for hot-spot mapping:** Rejoin LAT/LON from the raw table to enable point-density or cluster-based hot-spot maps in Power BI, which would sharpen the division-level findings in Section 5.4–5.5.
- **Secure database credentials:** Move the PostgreSQL connection string to environment variables or a secrets manager ahead of any shared deployment.
- **Extend the dashboard:** Add a year-over-year comparison view and an arrest/clearance-rate KPI (using Status Desc) to help evaluate case-outcome trends alongside incident volume.

10. Conclusion

This project delivered a complete analytics pipeline for the LAPD crime-incident dataset: a documented, reproducible cleaning process; a structured PostgreSQL data store preserving both raw and analysis-ready tables; a thorough exploratory analysis of temporal, spatial and demographic patterns; and a five-page interactive Power BI dashboard for ongoing self-service reporting.

The analysis confirms clear, actionable patterns — a pronounced midday and early-evening concentration of incidents, a small set of divisions driving a disproportionate share of citywide volume, and vehicle- and theft-related offenses as the leading crime categories — while also surfacing two data-quality artifacts (victim-age imputation and default timestamps) that should be corrected before the demographic and hourly findings are used for high-stakes decisions. Addressing the recommendations in Section 9, particularly the age-imputation fix and geospatial reintroduction, would meaningfully sharpen the precision of future iterations of this analysis.

Appendix A: Technical Stack Summary

Layer	Tool / Library
Data preparation & EDA	Python 3, pandas, NumPy
Visualization (notebook)	Matplotlib, Seaborn
Database	PostgreSQL
Database connectivity	SQLAlchemy, psycopg2-binary
Business Intelligence	Microsoft Power BI Desktop
Notebook environment	Jupyter Notebook (.ipynb)

Appendix B: Source Files

- Crime_data_Analysis.ipynb — Python notebook containing the full data-cleaning and EDA workflow.
- crime_data_analysis_dashboard.pbix — Power BI project file containing the five-page interactive dashboard described in Section 7.
- Crime_Data_from_2020_to_Present.csv — original source dataset (referenced by the notebook; not included in this report).